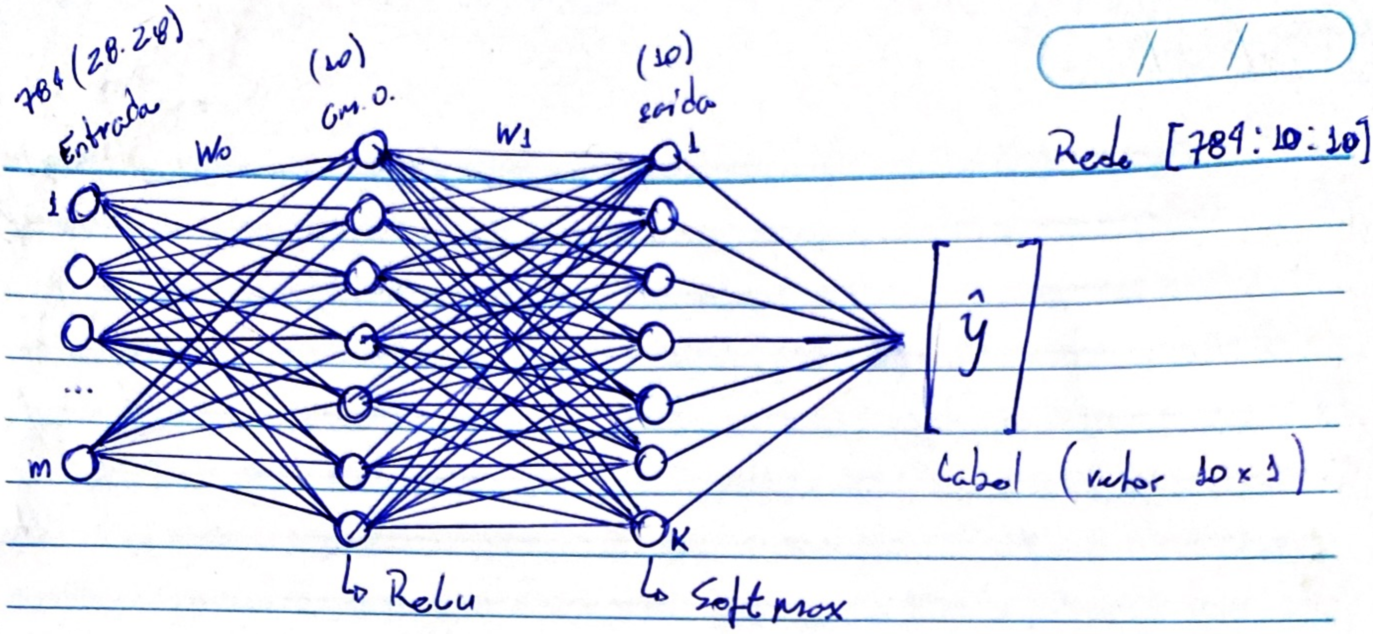
Rede [784:10:10]



A^0 = Entrada (X)

$z_1 = A^0 . W^0 + b^0$

$A^1 = Relu(z^1)$

$z^2 = A^1 . W^1 + b^1$

$A^2 = softmax(z^2)$

↓

Forward

$dz^2 = A^2 - \hat{y}$

$dW^1 = \left(\frac{1}{m}\right) \cdot dz^2 \cdot A^{1T}$

$db^1 = \left(\frac{1}{m}\right) \sum dz^2$

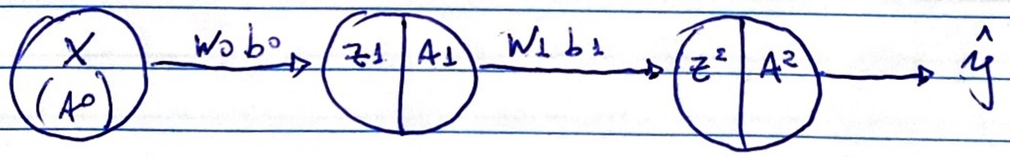
$dz^1 = W^{1T} \cdot dz^2 \odot d.Relu(z_1)$

$dW^0 = \left(\frac{1}{m}\right) . dz^1 . X^T$

$db^0 = \left(\frac{1}{m}\right) \sum dz^1$

↓

Back propagation



$Relu = max(0, z_1)$

$softmax = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$ → $e = 2,7182...$